My three priorities for impact in SLT/SLP research and teaching

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Land acknowledgement

I live in amiskwacîwâskahikan / Edmonton.

My three priorities

1. **Economic Impact:** I train resilient and future-proofed workers.
2. **Health Impact:** I develop, teach, and advocate for accessible clinical tools.
3. **Societal Impact:** I work to ensure health systems and research serve the entire population.

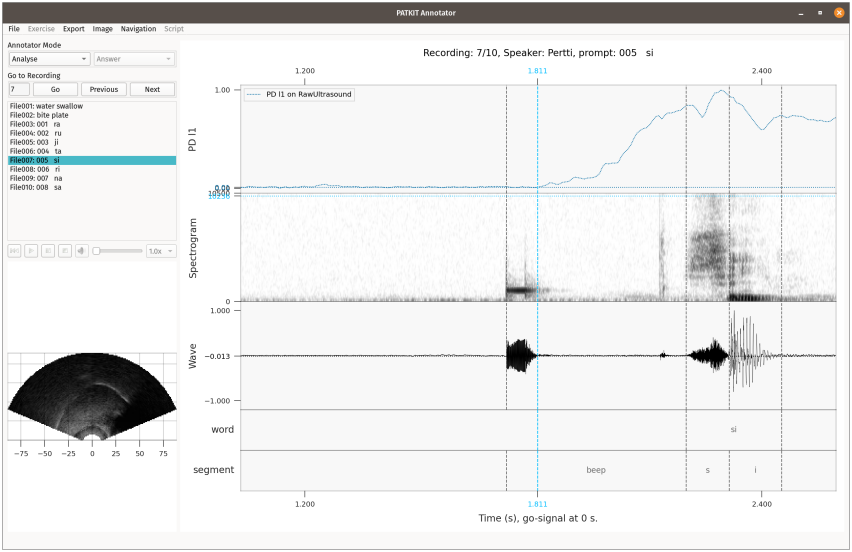
Priority 1: Training resilient and future-proofed workers

- I focus on foundational skills such as finding, critiquing, and curating information.
- I teach students how to learn with intention in general and especially how to efficiently learn and integrate new methods into their professional practice.
- I do not just tell of new tools, but I build the spaces and open source software (Palo 2024, Palo et al. 2026, Palo and Laporte 2026) for students to learn in practice and think critically about tongue ultrasound (Palo 2019), air flow (Palo and Cowen 2017), electroglottography (Waaramaa et al. 2014), etc.
- I want to provide joint courses with other programs to teach students practical collaboration with other professionals such as physicians, occupational therapists, physical therapists, and even engineers (University of Helsinki 2011-2012).

Priority 2: Accessible clinical tools

- Articulation provides a deeper understanding of speech production: It links neural processes and the produced speech.
- Ultrasound is my go-to method for articulation because it is relatively cheap, very portable, and intuitive for users who are knowledgeable in articulation and anatomy.
- It is useful in studying pathologies such as stuttering, Parkinsons disease, or covert errors (Lloyd et al. 2019), and it is an effective way of providing biofeedback.
- Example of speech analysis with PATKIT (Palo et al. 2026).
- Ultrasound video of silent speech articulation.

PLACE HOLDER SLIDE: PATKIT example



PLACE HOLDER SLIDE: ultrasound video example

Priority 3: Serving the entire population

- To connect academia with the general public I regularly contribute to a role-playing convention in Finland on the topics of speech articulation and responsible use of existing languages in fictional settings (Ropecons 2022-2025).
- For last two years I have been building relationships with Indigenous communities in and around Edmonton.
- Decolonisation and social justice work are necessary in ensuring our health systems and research are representative of all of humanity and not just some fraction of it (Lönnqvist et al. 2017).
- In Ireland I can see that my skills in community connection building can be useful in working with for example the The Gaeltacht, the Traveller community and migrant populations.

Concrete action on the priorities

1. With university and grant funding, I want to setup a world class teaching laboratory at University of Limerick. With my experience of designing and building such facilities (University of Helsinki 2020, University of Alberta 2024-2026), I can design it to fit in limited space or even make it possible to set the lab up for a class and quickly pack it away after.
2. I am planning an international project to study Parkinsons with ultrasound. Given Canada's recent joining of EU research programs, this will be a major EU grant with collaborators from Canada and Finland on this topic with methods I have developed over the past decade (Palo 2019, Palo et al. 2025; 2026).
3. To bring speech research and therapy to the whole population, I plan to take the idea of the mobile lab/clinic idea from the Netherlands (Rebernik et al. 2025) and adapt it to the Irish conditions and needs as a major grant funded project.

Thank you!

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